

Ashrith (Ash) Edukulla

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Education

University of Michigan, Ann Arbor — B.S.E. in Robotics | GPA 3.8/4.0 | Expected Graduation December 2027

Courses: Data Structures & Algorithms, Deep Learning, Computer Vision, Simultaneous Localization and Mapping & Navigation, Controls, Robot Operating Systems, Dynamics & Vibration, Intro to Electrical Circuits

Current Courses: Machine Learning, Intro to Reinforcement Learning, 3D Robot Perception, Autonomous Driving

Work Experience

NiFT – Co-Team Lead/Robotics ML Engineer

August 2026 – Present

- Co-lead the NiFT Autonomous Shuttle team, developing an end-to-end autonomous driving stack spanning planning, localization, control, and safety for multi-camera/LiDAR shuttles on NVIDIA AGX Thor.
- Lead deployment and real-time optimization of NVIDIA Alpamayo 1.5/2, accelerating VLA inference via TensorRT Edge-LLM, FP8/INT8 quantization, CUDA Graphs, KV-cache tuning, and zero-copy pipelines.
- Integrate Alpamayo into planning and control systems, linking multi-camera perception and 6.4 s waypoint trajectories to deterministic tracking, safety controls, and p50/p95/p99 latency profiling.

Moby Robotics – Perception Machine Learning Engineer

June 2026 – August 2026

- Built underwater perception systems for nodules, marine life, and open-set objects, improving detection performance by 30% over vanilla VLMs using real-world field data collected during at-sea deployments.
- Developed multi-camera sensor-fusion pipeline combining 360° imagery, stereo depth, and 3D reconstruction to improve spatial consistency for autonomous underwater mapping, navigation, and marine data capture.
- Profiled and optimized perception workloads on resource-constrained edge compute, balancing detection accuracy, inference latency, and memory usage for real-time deployment on embedded robotic systems.

Scalable Spatial Intelligence Lab – Undergraduate Researcher

March 2026 – Present

- Developed a latent world-model navigation system achieving 78% ObjectNav success on HM3D environments, improving generalization over prior baseline(OpenFrontier) through spatial memory and planning decisions.
- Implemented a long-horizon navigation stack combining visual observations, language-conditioned goals, episodic memory, and predictive world-model representations for robust sequential decision-making.
- Built Habitat evaluation pipelines to benchmark navigation agents across unseen environments, measuring success rate, path efficiency, and compute-performance tradeoffs for fair and reproducible comparison.

Field Robotics Group Lab – Undergraduate Researcher

December 2025 – May 2026

- Implemented hierarchical coverage planning using breadth first global search with locally biased rapidly-exploring random tree sampling for shipwreck inspection, improving exploration efficiency.
- Developed occupancy-grid and frontier-exploration algorithms to update coverage state and reduce redundant traversal under partial observability, improving decisions with noisy underwater sensor inputs.
- Built multimodal mapping pipeline fusing sonar and stereo vision into a unified spatial representation for autonomous inspection, improving robustness of 3D scene understanding across varying conditions.
- Designed state-driven switching between long-range sonar exploration and close-range stereo inspection, enabling adaptive sensing that balances global exploration with high-resolution local reconstruction.

EECS 467, ROB 330, ROB201, ROB 101 – Student Instructor

December 2024 – Present

- Mentored students in SLAM, Linear Algebra and Differential Equations for robotics, plus C++, Python, ROS software covering perception, SLAM, sensor fusion, planning, and controls for autonomous robotic systems
- Debugged end-to-end robot systems during labs and office hours, diagnosing software, sensor, and integration failures across full-stack autonomous robotics systems spanning HW/SW layers.

Papers

Multi Map Navigation World Action Models (Upcoming): First Author

Multimodal Sensor Fusion Shipwreck Coverage Planning (Upcoming): Co-First Author

OceanSim 2.0: Underwater Robotics Simulator (OCEANS 2026): Third Author

Skills

Languages & Systems: C++, Python, Linux, Git, ROS 2, MATLAB, Julia

ML/CV: PyTorch, NumPy, Stereo Vision, Visual Odometry, Multi-View Stereo, 3D Gaussian Splatting

Robotic & 3D: SLAM, Sensor Fusion, State Estimation, Path Planning, Point Cloud Processing, 3D Reconstruction